

PyOmniGraph - Getting and hosting your own federated copies of subgraphs of Wikidata and other knowledge graphs^{*}

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Abstract

Wikidata is an example for a very large, crowd sourced general knowledge graph backed by a world wide community since 2012. Factgrid is using the same Wikibase infrastructure as Wikidata since 2017 for historical research projects. GOV (Geschichtliches/Genealogisches Ortsverzeichnis) is an online gazetteer for searching ancient places. Wikidata grew so large that the Blazegraph endpoint powering it was pushed to the 4 Terrabyte limit and the Wikimedia Foundation therefore decided to split the knowledge graph to host two instances. The challenges of federated queries that came up in the above use cases led to the need of being able to work with subgraphs of the RDF graphs and try out different alternative Endpoints such as QLever and Virtuoso and others. While typical research projects that do such subgraphing and benchmarks roll their own environments we decided to provide a python based Open Source library pyomnigraph to provide a unified infrastructure with a standard set of REST APIs and CLI commands to improve the handling and potentially get a bigger matrix of subgraphs versus endpoints to work with. In this work we report on pyomnigraph usage on a matrix of three RDF graphs: Wikidata, Factgrid and GOV and endpoints blazegraph, jena, Qlever on a selection of subgraph combinations. We show that typical frustrations of scaling multi-hop federated queries can be avoided if the subgraphs are locally loaded and federation is applied locally.

Keywords

Wikidata, FactGrid, QLever, Blazegraph, Apache Jena, GOV, RDF triple store, federated SPARQL, benchmark, PyOmniGraph

1. Introduction

Wikidata [1] started in 2012 as an effort to back the different international Wikipedia sites with a common set of facts. Initially, about 2 million entities were harvested from existing Wikipedia content. As of 2016, over 100 million entities have been curated by a growing worldwide community of contributors. In 2022, we reported on How to get and host your own copy of Wikidata [2]. For quite a few use case the need for federated queries based on subgraphs of diverse public and internal knowledge-graphs has been our motivation to try out different combinations of endpoints and subgraphs. In the case of Wikidata we even have been forced to do so by the 2025 graphsplitted [3].

Public alternatives to the official Wikidata Query Service WDQS are still rare. Our attempts to convince the Wikimedia Foundation to support a public Mirror infrastructure have so far been in vain. Table 1 lists the seven workin public Wikidata endpoints currently configured in <https://nicescholia.wikidata.dbis.rwth-aachen.de/api/endpointsnicescholia> to be watched as potential alternative scholia backends.

The Wikimedia Foundation itself now serves three: the <https://query-legacy-full.wikidata.org/sparqllegacy> full graph and, since the WDQS graph split, a [---

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Table 1

Public Wikidata query endpoints

Provider	Triple Store	Triples (B)	Last update
Wikimedia Foundation (legacy full)	Blazegraph	22.0	continuous
Wikimedia Foundation (main)	Blazegraph	8.6	continuous
Wikimedia Foundation (scholarly)	Blazegraph	8.7	continuous
RWTH Aachen DBIS	Blazegraph	17.0	2025-06-01
RWTH Aachen DBIS	QLever	–	continuous
University of Freiburg	QLever	–	continuous
OpenLink Software (truthy)	Virtuoso	–	2025-11-01

`main.wikidata.org/sparqlmain` and a `https://query-scholarly.wikidata.org/sparqlscholarly` half. RWTH Aachen DBIS hosts a full `https://wdqs.wikidata.dbis.rwth-aachen.de/sparqlBlazegraph` mirror alongside a `https://qllever-api.wikidata.dbis.rwth-aachen.de/QLever` endpoint; the University of Freiburg `https://qllever.dev/api/wikidataQLever` service has moved to the `qllever.dev` host. `https://truthy.demo.openlinksw.com/sparql` OpenLink Software provides a Virtuoso instance restricted to truthy statements. These alternatives still have their own limitations regarding SPARQL language coverage, data currency and completeness.

2. Related Work

Willighagen et. al. [3] report on the three options that have been considered for surviving the 2025 Wikidata graph split: Federated queries between the main and the scholar graph, using qllever with modified queries on a full graph that does not suffer from the 4 TB limit of the blazegraph endpoint or using snapquery with named parameterized queries as a middle ware to separate the endpoints concerns from the domain needs.

CONSTRUCT queries in SPARQL have been studied by Kostylev et al. [4].

3. The pyomnigraph Resource

pyomnigraph provides a unified python interface for multiple graph databases. The software is open source under Apache 2.0 license and hosted on github.

3.1. Supported Backends

3.2. Command-Line Interfaces

There are two command line interfaces available for pyomnigraph: `omnigraph` for server management and `rdfdump` for data operations.

4. Use cases and Data Sources

The use cases for our experiments are motivated by the eat your own dog food approach.

5. Federated Query Benchmark

6. Evaluation

7. Availability

8. Conclusion and Future Work

Declaration on Generative AI

AI assistance in this project is limited to spelling, grammar, and formatting/build tooling; AI is NOT used to generate scientific content, arguments, or prose.

References

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